

Truss-by-truss design and calculation review

Revision 3: connected model with W-section trolley rails. This package gives an explicit candidate geometry and member schedule for T-S, T1, T-E, T-W and T-N. It traces loads from the framing plan into each truss, records the controlling member checks, and tests lighter alternatives. It remains a preliminary first-order gravity design study, not a fabrication or construction design.

The model has a feasible local result under its selected screens: **7,495 lb** of primary steel plus a 10% connection-weight allowance, a highest strength screening ratio of **0.899**, and a largest service vertical movement of **0.730 inch**. This is approximately 1,738 lb (19%) below the prior 9,233 lb HSS study. Separating upper and lower chords, changing B2 end behavior and using W rails all changed this comparison; the savings cannot be attributed to one change alone.

Truss	Lower chord	Upper chord(s)	Webs / posts
T-S	HSS 2 x 2 x 1/8	HSS 2 x 2 x 1/8	HSS 2 x 2 x 1/8
T1	W8X24	HSS 2 x 2 x 1/8	HSS 2.5 x 2.5 x 1/8
T-E	HSS 6 x 6 x 1/4	HSS 2.5 x 2.5 x 1/8	HSS 2.5 x 2.5 x 1/8
T-W	HSS 6 x 6 x 1/4	HSS 6 x 6 x 1/4	HSS 3 x 3 x 3/16
T-N	HSS 4 x 4 x 3/16	HSS 3 x 3 x 3/16	HSS 3 x 3 x 1/8

B2 is W8X31, 8.00 inches actual depth. T1 remains a full truss with a W8X24 bottom chord, 7.93 inches deep. W8X24 means nominal 8-inch depth and 24 lb/ft; the 24 is not a wall thickness. Both rails need actual trolley flange compatibility and local wheel-load checks before any lifting rating can be assigned.

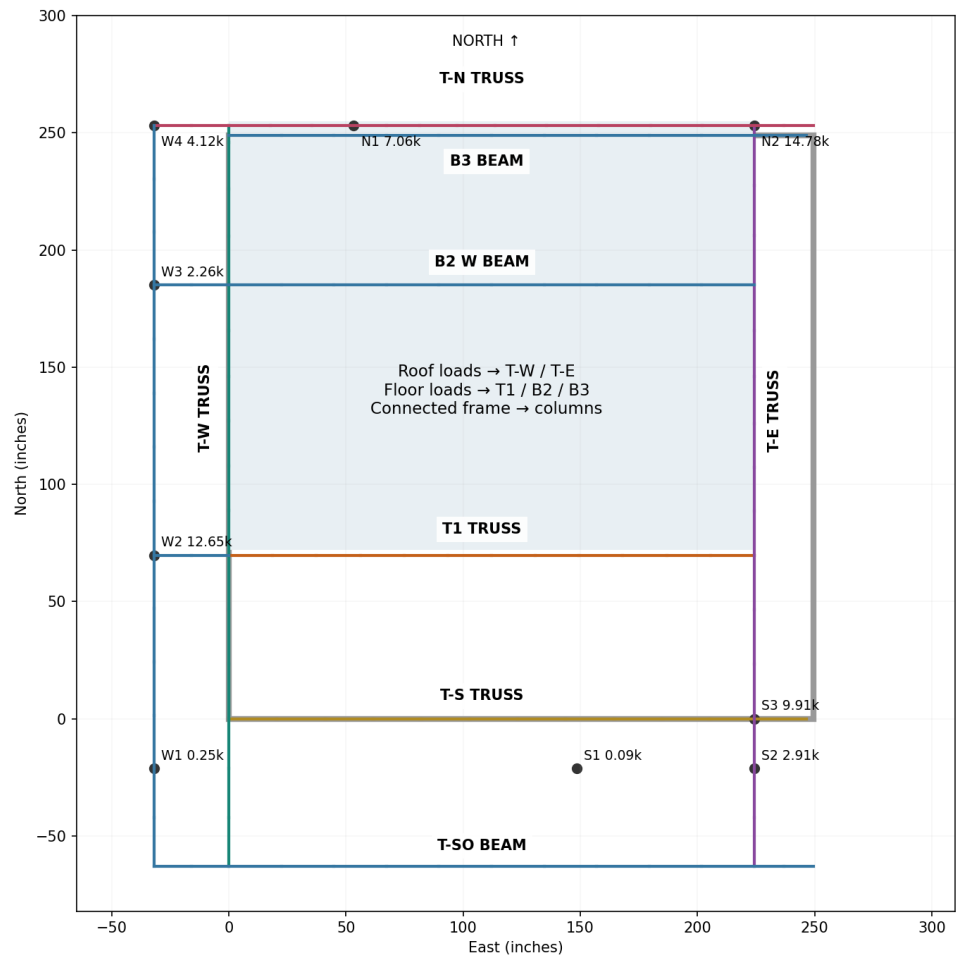
The small T-S members are at the bottom of the tested catalogue, not proven absolute minima. Their low demand depends on roof purlins spanning east-west to T-W/T-E. Wind, facade detailing and connection fabrication can set substantially larger minimum sizes. All schedules in this report are conditional candidates.

What this delivers, and what remains

Delivered: truss geometry with member IDs, load-path actions, trial sizes, compression/bending screens, moving-hoist gravity cases and a recorded lighter-section search. Remaining: site lateral loads and stability, bracing design, joint design, trolley local flange effects, cap collectors, column bases and foundations. These unresolved checks can invalidate the light sections shown here.

1. Top-down load path

Load-path plan • revised W-rail candidate



Column labels: service-storage vertical reactions in kip (1 kip = 1,000 lb).
Roof and floor are applied once; inter-truss transfers are internal forces, not additional building weight.

Blue horizontal members are beams. Each colored perimeter line represents the full-height truss on its individual elevation sheet. No added gravity bearing is assigned to the existing walls, slab or cabinets. S1 is modeled but still has no upper-frame connection. S3 is assumed present.

2. Loads enter once, then transfer between members

Roof purlins are assumed to span east-west onto T-W and T-E. Floor joists span north-south onto T1, B2 and B3. The connected stiffness model then redistributes the reactions through the trusses, transfer beams and columns. A load transferred from B2 into T-E is not added again as building weight.

Group	Direct service load, lb	Sum of vertical actions from connected frame, lb
T-S	566	566
T1	7,416	7,416
B2	11,856	11,856
B3	7,530	7,530
T-E	11,609	11,609
T-W	11,209	11,209
T-N	2,072	2,072

Direct load includes the assigned roof/floor/wall load and the member group self-weight, for D + perimeter-storage live load + roof live load. Positive connection action is upward on the named group. Connection actions can include both upward and downward forces at different joints. The equality above is a group equilibrium check, not a statement that each truss carries the entire building.

Load cases and trolley assumptions

The inherited floor is 284.98 sq ft, with 125 psf live load in 36-inch north/east/west storage bands and 40 psf elsewhere. Roof live load is 20 psf on horizontal projection. Dead allowances are 9 psf floor, 9.5 psf actual solar roof surface, 6.6 psf cap surface, 4 psf gross upper walls, and the balcony/guard allowances from revision 2. The new frame self-weight is recalculated for every trial.

A single operating trolley is tested on either B2 or T1 at nine positions, from $x = 0$ to 224.25 inches. Each position adds 1,350 lb downward: the confirmed 1,000 lb lifted load, a provisional 25% dynamic increment, and an assumed 100 lb trolley/hoist. This is a sampled position envelope, not a continuous moving-load proof. The actual equipment weight, wheel spacing, flange contact and dynamic requirements must replace these assumptions. Simultaneous use of two hoists is not included.

Service = D + storage + roof live, with and without each trolley position. Strength = $1.2D + 1.6$ storage + 0.5 roof live, with trolley gravity factored 1.6; also $1.2D +$ storage + 1.6 roof live without trolley. These are selected study combinations, not the full legal design envelope. Full-area 125 psf and lateral cases from revision 2 are not requalified by this revision.

Movement criterion

The search holds maximum absolute vertical movement at any model node to 0.75 inch. This is a study-wide tolerance, not an L/360 certification for every beam or a hoist manufacturer alignment limit. Maximum movement with the trolley is 0.730 inch; the no-trolley storage case is 0.726 inch. Member-relative deflection, floor vibration and machinery tolerances still need review.

3. How the member sizes are found

Each truss is kept as a complete connected assembly. Upper chords, lower chords and webs are sized as separate families, while the global model retains their common nodes. Changing a family changes both stiffness and self-weight, so the entire frame is solved again after every trial. The geometry itself is held fixed; this run is a section search, not a truss-topology optimization.

Start with the earlier feasible HSS study, replace B2 and the T1 lower chord with real W-section candidates, and release B2 end bending rotations to represent shear seats. B2 end torsion remains restrained and needs a real detail.

Increase families that fail the selected combined-force screen. Then test every lighter catalogue section in each family and accept the first one that keeps the whole frame within the strength and movement screens.

Repeat complete sweeps until no family can be reduced. Three sweeps were run; the final sweep accepted zero reductions. This is a coordinate-wise local stopping point. Simultaneous section swaps, different bracing or different panel geometry can produce a different result.

Strength calculations used

Steel $E = 29,000$ ksi and specified $F_y = 50$ ksi. Compression uses $r = \sqrt{I_{\min}/A}$, $F_e = \pi^2 E / (KL/r)^2$, and $F_{cr} = F_y (0.658)^{(F_y/F_e)}$ when $F_y/F_e \leq 2.25$, otherwise $0.877 F_e$. Available compression $P_c = 0.9 F_{cr} A$; gross-section tensile yield $P_t = 0.9 F_y A$. Net-section fracture and connection rupture are not checked.

HSS elastic bending screens use $0.9 F_y S$ about each axis. W major-axis bending additionally uses AISC F2 lateral-torsional buckling with $C_b = 1.0$, capped at elastic yield. Slightly noncompact W flanges are conservatively capped at $0.9 \times 0.7 F_y S$ rather than credited with plastic capacity. This is deliberately a screening implementation, not every AISC limit state.

Combined metric = $\max(N_{\text{compression}}/P_c, N_{\text{tension}}/P_t) + |M_{\text{major}}|/M_{c\text{-major}} + |M_{\text{minor}}|/M_{c\text{-minor}}$. A separate approximate shear-plus-torsion ratio is checked, and the larger value controls. The search limit is 0.90. This conservative linear metric is not a full AISC H1/H3 code check. All force values are first-order; no second-order stability reduction has been included.

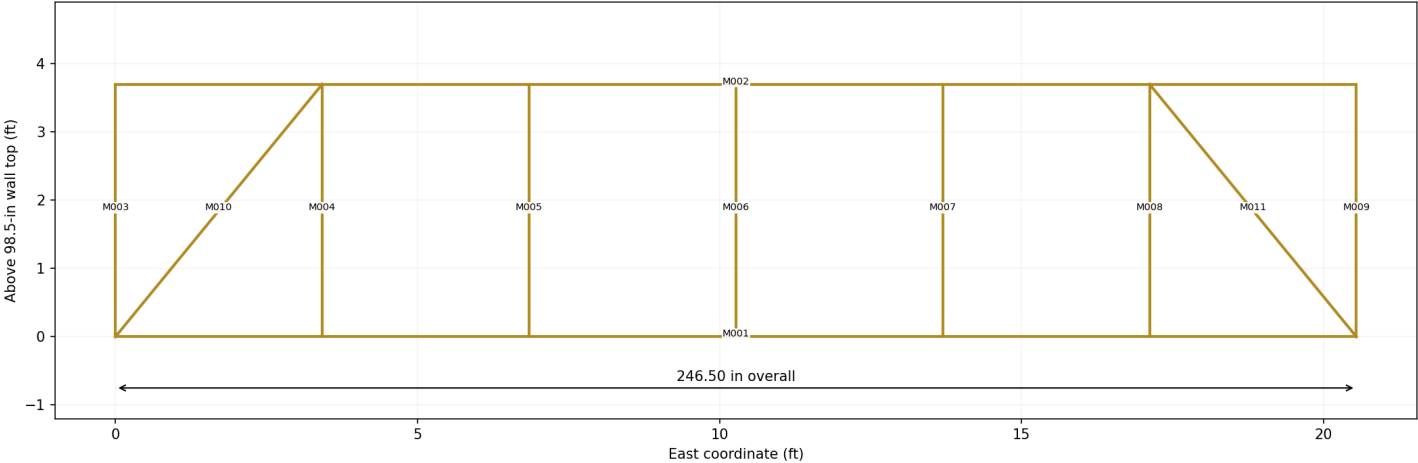
Required restraint assumptions: $K = 1$; chords and W members have effective lateral/torsional brace spacing at most 72 inches, limited by their original member length; web members use their original full length; columns use 98.5 inches. Numerical subdivisions never shorten the assumed buckling length. Those brace points and their force/stiffness requirements have not been designed. W torsion uses Saint-Venant stiffness only; restrained warping and flange distortion remain unresolved.

Closed-form solver tests cover cantilever bending, axial extension, torsion, simple-span point loading, end releases, and distinct strong/weak W-section axes. Load-path recovery checks each member group equilibrium. These verify the calculation implementation at this level, not the physical assumptions.

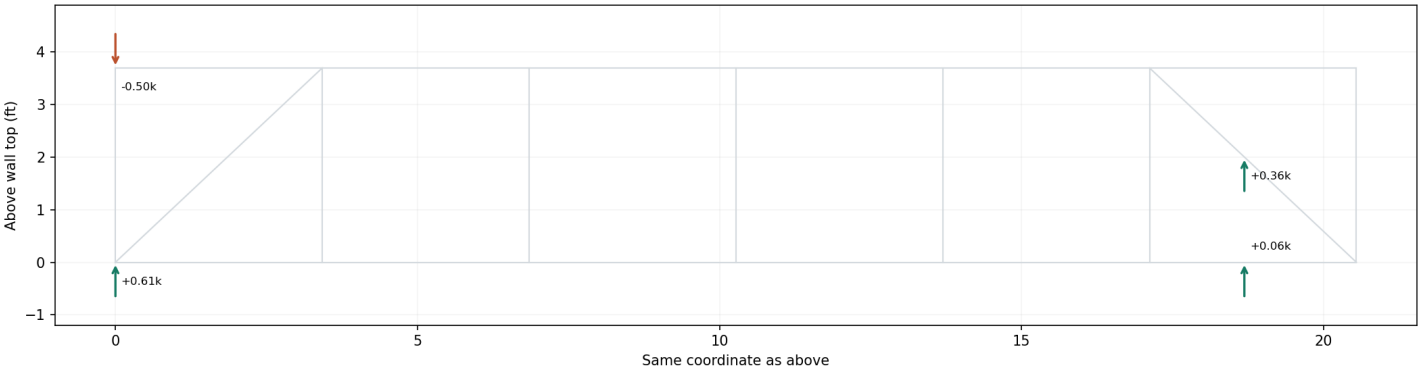
T-S — geometry and load transfer

lower: HSS2x2x0.125 | upper: HSS2x2x0.125 | web: HSS2x2x0.125

T-S — full truss, with physical member IDs



Service storage: 0.57 kip directly applied to this group; arrows = actions from connected framing



Overall analytical span 246.50 in; maximum depth above wall-top line 44.37 in. Physical member IDs match the force table. Member lengths are analytical centerline lengths, not cut lengths. Joints, offsets and weld gaps remain to be detailed.

Role	Candidate	Governing screen	Case
lower	HSS 2 x 2 x 1/8	0.212	strength T1@2/8
upper	HSS 2 x 2 x 1/8	0.244	strength T1@2/8
web	HSS 2 x 2 x 1/8	0.272	strength T1@2/8

The bottom arrows show only vertical interaction with adjacent framing in the no-trolley service case. Distributed direct loads are summarized in the title. Horizontal actions and joint moments also exist; connection-actions.csv contains all six components. A connection cannot be designed from the vertical arrows alone.

T-S — member force and capacity checks

ID / role	Length in	Comp. / tens. kip	Major / minor M kip-ft	Ratio
M001 / lower	246.5	0.09 / 0.00	0.36 / 0.02	0.212
M002 / upper	246.5	0.25 / 0.00	0.40 / 0.02	0.244
M003 / web	44.4	0.77 / 0.00	0.03 / 0.11	0.101
M004 / web	44.4	0.00 / 0.08	0.00 / 0.08	0.045
M005 / web	44.4	0.00 / 0.03	0.00 / 0.21	0.115
M006 / web	44.4	0.00 / 0.03	0.00 / 0.37	0.205
M007 / web	44.4	0.00 / 0.05	0.00 / 0.49	0.272
M008 / web	44.4	0.00 / 0.18	0.02 / 0.24	0.152
M009 / web	44.4	0.00 / 0.11	0.04 / 0.07	0.059
M010 / web	60.5	0.15 / 0.00	0.08 / 0.01	0.057
M011 / web	60.5	0.67 / 0.00	0.04 / 0.05	0.082

Each row reports the controlling sampled load case and finite-element segment within that physical member. Axial and moment components conservatively take the larger end value within that segment; they are not a single connection force vector. The CSV records the governing case, effective buckling length, capacities, shear and torsion for every row.

Worked check: governing member in this truss

M007: HSS 2 x 2 x 1/8; governing case strength T1@2/8. Assumed effective buckling length 44.4 in. Available compression 29.47 kip, tensile yield 37.80 kip, major bending 1.82 kip-ft, minor bending 1.82 kip-ft.

Axial term = 0.001; major-bending term = 0.001; minor-bending term = 0.270. Sum = 0.272. Separate shear/torsion screen = 0.022. Governing metric = 0.272, against the study target 0.900. This conditional screen does not include joint capacity or second-order effects.

T-S — why not make it lighter?

Family	Lighter trial	Saved lb	Global ratio	Global movement in
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Each row changes only that family and then re-solves the entire connected building. The illustrated trial is the next lighter catalogue entry by weight; the full search also tested other lighter entries. A ratio above 0.900 or movement above 0.750 inch rejects the trial. A rejection can be caused by a different member because stiffness changes redistribute loads.

A missing role is already at the lightest candidate in the tested catalogue. That is a catalogue floor, not proof of a physical minimum and not a fabrication recommendation.

Selected T-S weight including 10% connection allowance: 258 lb.

Connection actions to carry forward

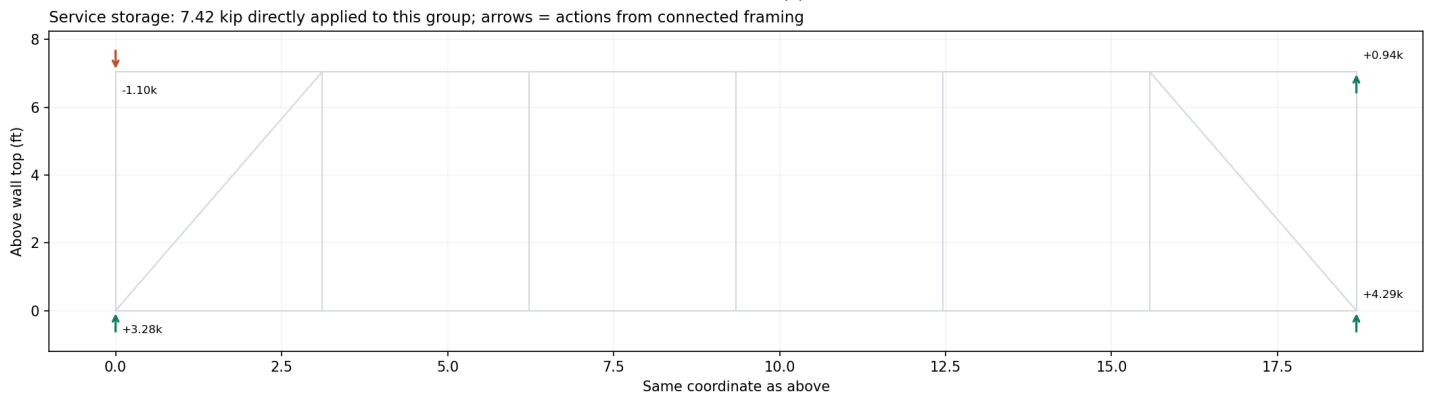
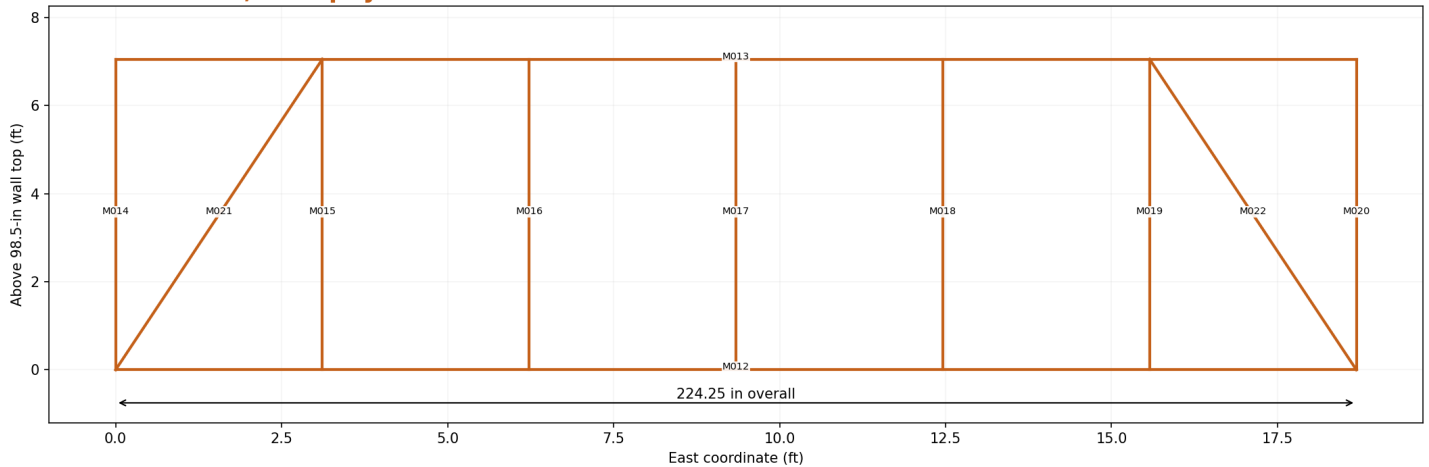
Node	Connected group(s)	Vertical lb	Moment magnitude kip-ft
0	T-W	+612	0.22
11	S3, T-E	+56	0.18
13	T-W	-500	0.02
24	T-E	+40	0.14
36	T-E	+358	0.07

These are service connection actions, included to expose the assumed transfer path. They are not a factored connection design schedule. The moment magnitude is for orientation only; connection design must use the signed component vectors and all governing combinations. Welds, gussets, bolts, HSS face yielding and local distortions remain unqualified.

T1 — geometry and load transfer

lower: W8X24 | upper: HSS2x2x0.125 | web: HSS2.5x2.5x0.125

T1 — full truss, with physical member IDs



Overall analytical span 224.25 in; maximum depth above wall-top line 84.64 in. Physical member IDs match the force table. Member lengths are analytical centerline lengths, not cut lengths. Joints, offsets and weld gaps remain to be detailed.

Role	Candidate	Governing screen	Case
lower	W8X24	0.446	strength T1@0/8
upper	HSS 2 x 2 x 1/8	0.496	strength T1@4/8
web	HSS 2.5 x 2.5 x 1/8	0.633	strength T1@3/8

The bottom arrows show only vertical interaction with adjacent framing in the no-trolley service case. Distributed direct loads are summarized in the title. Horizontal actions and joint moments also exist; connection-actions.csv contains all six components. A connection cannot be designed from the vertical arrows alone.

T1 retains its full 84.64-inch truss depth. The W8X24 is only the lower chord/trolley rail. Open middle bays require moment-resisting joints; their lower chord is not assumed to act alone as a full-span simply supported beam.

T1 — member force and capacity checks

ID / role	Length in	Comp. / tens. kip	Major / minor M kip-ft	Ratio
M012 / lower	224.2	0.00 / 4.28	33.70 / 0.06	0.446
M013 / upper	224.2	4.90 / 0.00	0.44 / 0.01	0.496
M014 / web	84.6	1.74 / 0.00	0.02 / 0.40	0.204
M015 / web	84.6	0.00 / 10.98	0.00 / 0.06	0.249
M016 / web	84.6	0.00 / 0.12	0.00 / 0.27	0.094
M017 / web	84.6	0.04 / 0.00	0.00 / 0.53	0.179
M018 / web	84.6	0.00 / 0.05	0.00 / 0.62	0.208
M019 / web	84.6	0.00 / 10.24	0.00 / 0.37	0.336
M020 / web	84.6	0.00 / 1.70	0.01 / 0.15	0.089
M021 / web	92.5	12.07 / 0.00	0.02 / 0.41	0.633
M022 / web	92.5	11.51 / 0.00	0.03 / 0.17	0.534

Each row reports the controlling sampled load case and finite-element segment within that physical member. Axial and moment components conservatively take the larger end value within that segment; they are not a single connection force vector. The CSV records the governing case, effective buckling length, capacities, shear and torsion for every row.

Worked check: governing member in this truss

M021: HSS 2.5 x 2.5 x 1/8; governing case strength T1@3/8. Assumed effective buckling length 92.5 in. Available compression 24.61 kip, tensile yield 48.15 kip, major bending 2.99 kip-ft, minor bending 2.99 kip-ft.

Axial term = 0.490; major-bending term = 0.007; minor-bending term = 0.136. Sum = 0.633. Separate shear/torsion screen = 0.006. Governing metric = 0.633, against the study target 0.900. This conditional screen does not include joint capacity or second-order effects.

T1 — why not make it lighter?

Family	Lighter trial	Saved lb	Global ratio	Global movement in
lower	W6X20	82.2	0.919	0.732
web	HSS 2 x 2 x 1/8	60.6	1.024	0.731

Each row changes only that family and then re-solves the entire connected building. The illustrated trial is the next lighter catalogue entry by weight; the full search also tested other lighter entries. A ratio above 0.900 or movement above 0.750 inch rejects the trial. A rejection can be caused by a different member because stiffness changes redistribute loads.

A missing role is already at the lightest candidate in the tested catalogue. That is a catalogue floor, not proof of a physical minimum and not a fabrication recommendation.

Selected T1 weight including 10% connection allowance: 834 lb.

Connection actions to carry forward

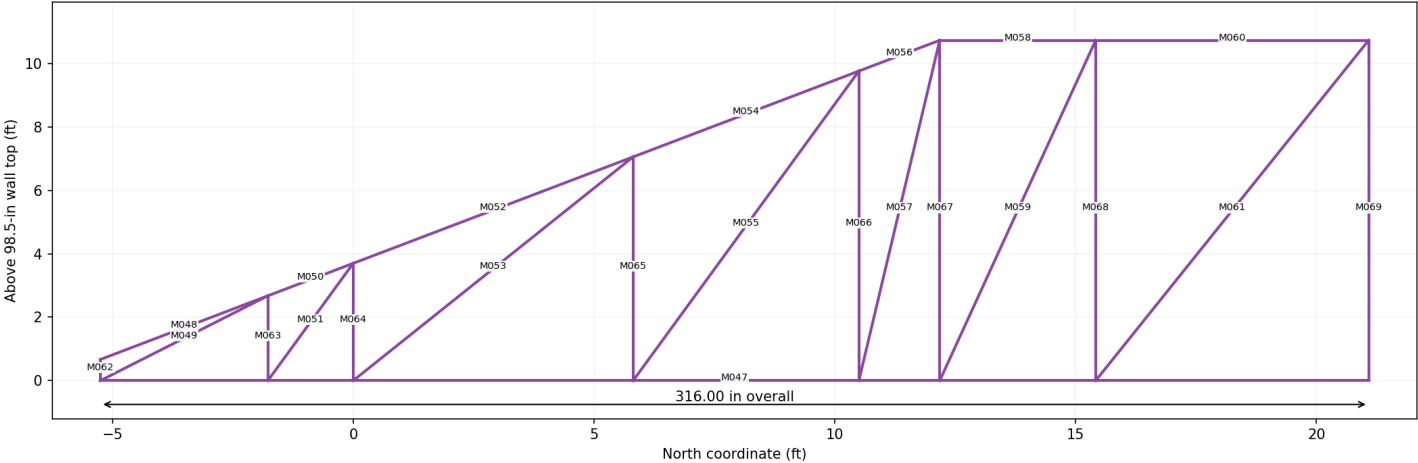
Node	Connected group(s)	Vertical lb	Moment magnitude kip-ft
38	O2, T-W	+3,277	27.10
50	T-E	+4,289	1.05
51	T-W	-1,095	0.08
63	T-E	+944	0.06

These are service connection actions, included to expose the assumed transfer path. They are not a factored connection design schedule. The moment magnitude is for orientation only; connection design must use the signed component vectors and all governing combinations. Welds, gussets, bolts, HSS face yielding and local distortions remain unqualified.

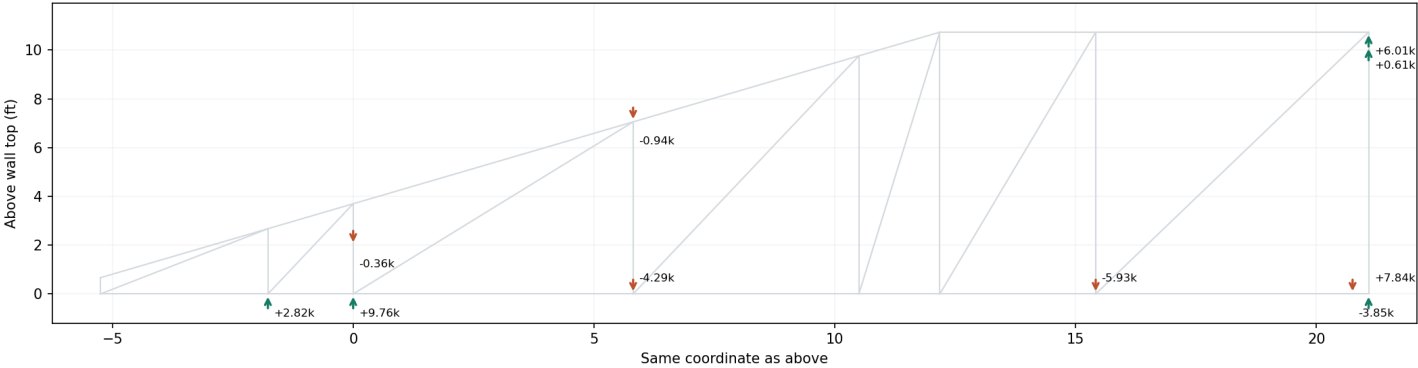
T-E — geometry and load transfer

lower: HSS6x6x0.25 | upper: HSS2.5x2.5x0.125 | web: HSS2.5x2.5x0.125

T-E — full truss, with physical member IDs



Service storage: 11.61 kip directly applied to this group; arrows = actions from connected framing



Overall analytical span 316.00 in; maximum depth above wall-top line 128.75 in. Physical member IDs match the force table. Member lengths are analytical centerline lengths, not cut lengths. Joints, offsets and weld gaps remain to be detailed.

Role	Candidate	Governing screen	Case
lower	HSS 6 x 6 x 1/4	0.784	strength B2@0/8
upper	HSS 2.5 x 2.5 x 1/8	0.664	strength roof
web	HSS 2.5 x 2.5 x 1/8	0.708	strength T1@8/8

The bottom arrows show only vertical interaction with adjacent framing in the no-trolley service case. Distributed direct loads are summarized in the title. Horizontal actions and joint moments also exist; connection-actions.csv contains all six components. A connection cannot be designed from the vertical arrows alone.

T-E — member force and capacity checks

ID / role	Length in	Comp. / tens. kip	Major / minor M kip-ft	Ratio
M047 / lower	316.0	0.05 / 0.00	1.86 / 0.43	0.784
M048 / upper	48.2	0.00 / 0.13	0.46 / 0.03	0.168
M049 / web	52.7	0.00 / 1.19	0.11 / 0.04	0.072
M050 / upper	24.5	0.00 / 1.23	0.52 / 0.03	0.206
M051 / web	49.2	3.46 / 0.00	0.33 / 0.04	0.210
M052 / upper	80.5	1.71 / 0.00	1.43 / 0.06	0.549
M053 / web	109.7	12.65 / 0.00	0.07 / 0.03	0.708
M054 / upper	65.0	9.97 / 0.00	1.03 / 0.10	0.664
M055 / web	129.9	0.00 / 4.13	0.10 / 0.02	0.128
M056 / upper	23.3	8.65 / 0.00	0.31 / 0.06	0.312
M057 / web	130.3	0.00 / 3.76	0.10 / 0.06	0.131
M058 / upper	38.9	6.47 / 0.00	0.73 / 0.04	0.408
M059 / web	134.5	0.00 / 2.22	0.05 / 0.07	0.086
M060 / upper	68.0	5.78 / 0.00	1.26 / 0.01	0.597
M061 / web	145.6	0.00 / 14.21	0.07 / 0.20	0.386
M062 / web	8.0	0.98 / 0.00	0.35 / 0.02	0.142
M063 / web	32.1	1.33 / 0.00	0.17 / 0.06	0.108
M064 / web	44.4	0.28 / 0.00	0.49 / 0.24	0.251
M065 / web	84.6	0.00 / 1.70	0.01 / 0.15	0.089
M066 / web	117.1	3.34 / 0.00	0.18 / 0.08	0.288
M067 / web	128.8	0.83 / 0.00	0.08 / 0.05	0.106
M068 / web	128.8	4.75 / 0.00	0.03 / 0.11	0.394
M069 / web	128.8	4.52 / 0.00	0.18 / 0.56	0.580

Each row reports the controlling sampled load case and finite-element segment within that physical member. Axial and moment components conservatively take the larger end value within that segment; they are not a single connection force vector. The CSV records the governing case, effective buckling length, capacities, shear and torsion for every row.

Worked check: governing member in this truss

M047: HSS 6 x 6 x 1/4; governing case strength B2@0/8. Assumed effective buckling length 72.0 in. Available compression 219.98 kip, tensile yield 235.80 kip, major bending 35.74 kip-ft, minor bending 35.74 kip-ft.

Axial term = 0.000; major-bending term = 0.052; minor-bending term = 0.012. Sum = 0.064. Separate shear/torsion screen = 0.784. Governing metric = 0.784, against the study target 0.900. This conditional screen does not include joint capacity or second-order effects.

T-E — why not make it lighter?

Family	Lighter trial	Saved lb	Global ratio	Global movement in
lower	HSS 4 x 4 x 3/8	50.7	1.160	0.767
upper	HSS 2 x 2 x 1/8	27.1	1.019	0.731
web	HSS 2 x 2 x 1/8	111.0	1.370	0.733

Each row changes only that family and then re-solves the entire connected building. The illustrated trial is the next lighter catalogue entry by weight; the full search also tested other lighter entries. A ratio above 0.900 or movement above 0.750 inch rejects the trial. A rejection can be caused by a different member because stiffness changes redistribute loads.

Selected T-E weight including 10% connection allowance: 1,185 lb.

Connection actions to carry forward

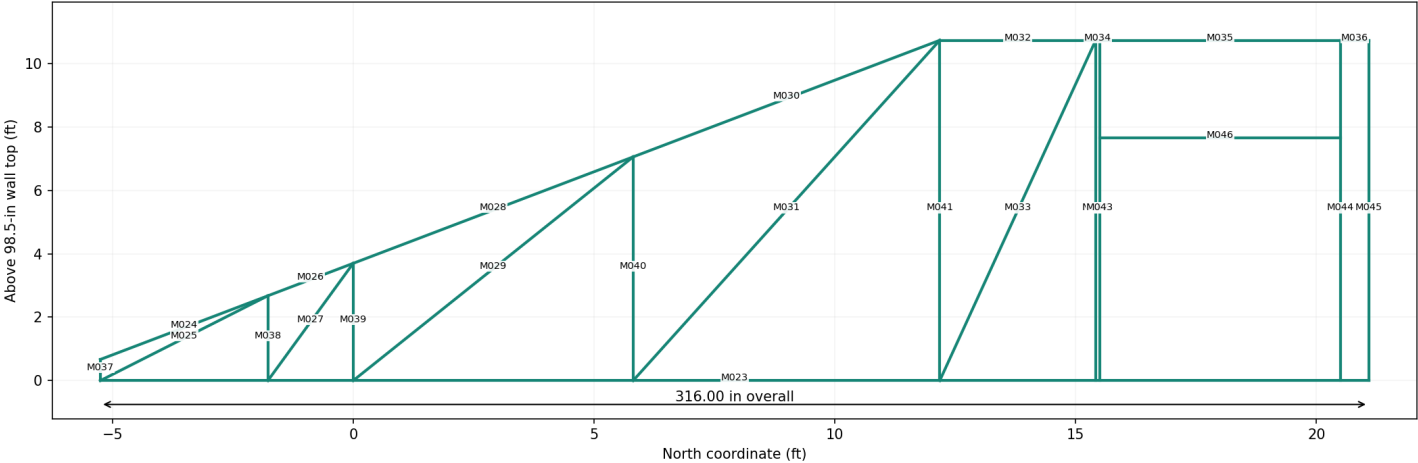
Node	Connected group(s)	Vertical lb	Moment magnitude kip-ft
11	S3, T-S	+9,764	0.70
24	T-S	-40	0.14
36	T-S	-358	0.07
50	T1	-4,289	1.05
63	T1	-944	0.06
175	T-SO	-47	0.16
177	S2	+2,821	0.56
185	B2	-5,928	0.00
188	B3	-3,846	10.42
189	N2, T-N	+7,842	8.44
229	T-N	+6,012	0.54
256	T-N	+11	0.14
257	T-N	+612	0.03

These are service connection actions, included to expose the assumed transfer path. They are not a factored connection design schedule. The moment magnitude is for orientation only; connection design must use the signed component vectors and all governing combinations. Welds, gussets, bolts, HSS face yielding and local distortions remain unqualified.

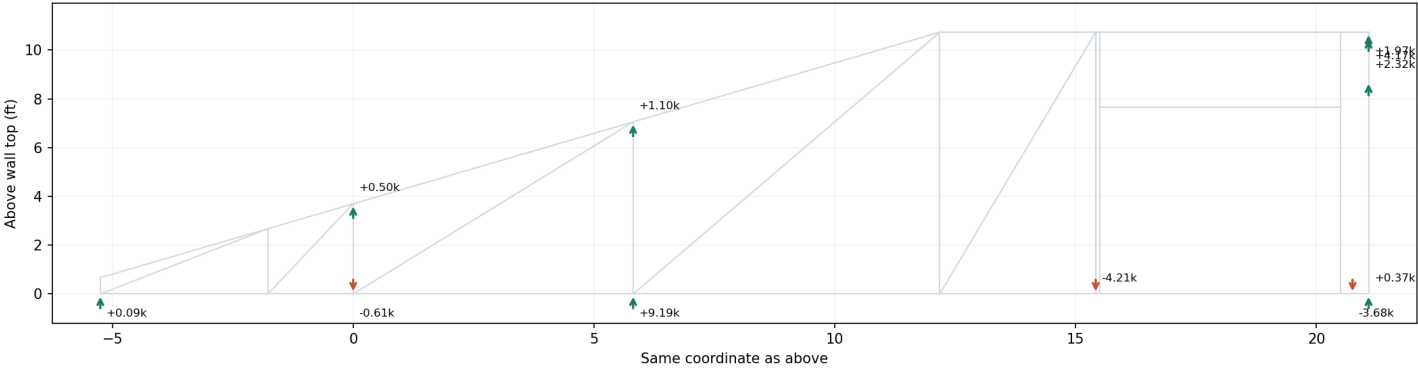
T-W — geometry and load transfer

lower: HSS6x6x0.25 | upper: HSS6x6x0.25 | web: HSS3x3x0.188

T-W — full truss, with physical member IDs



Service storage: 11.21 kip directly applied to this group; arrows = actions from connected framing



Overall analytical span 316.00 in; maximum depth above wall-top line 128.75 in. Physical member IDs match the force table. Member lengths are analytical centerline lengths, not cut lengths. Joints, offsets and weld gaps remain to be detailed.

Role	Candidate	Governing screen	Case
lower	HSS 6 x 6 x 1/4	0.778	strength B2@0/8
upper	HSS 6 x 6 x 1/4	0.334	strength B2@0/8
web	HSS 3 x 3 x 3/16	0.609	strength B2@0/8

The bottom arrows show only vertical interaction with adjacent framing in the no-trolley service case. Distributed direct loads are summarized in the title. Horizontal actions and joint moments also exist; connection-actions.csv contains all six components. A connection cannot be designed from the vertical arrows alone.

The rear opening is deliberately untriangulated. Its frame action helps explain why T-W retains a much larger upper chord than T-E. Do not add diagonals through the balcony/door opening without an architectural revision.

T-W — member force and capacity checks

ID / role	Length in	Comp. / tens. kip	Major / minor M kip-ft	Ratio
M023 / lower	316.0	0.00 / 0.64	3.36 / 0.53	0.778
M024 / upper	48.2	0.00 / 0.45	0.56 / 0.22	0.024
M025 / web	52.7	0.00 / 0.77	0.03 / 0.12	0.033
M026 / upper	24.5	0.00 / 1.45	0.37 / 0.26	0.024
M027 / web	49.2	0.00 / 1.45	0.05 / 0.29	0.071
M028 / upper	80.5	0.00 / 2.75	2.38 / 0.83	0.101
M029 / web	109.7	0.00 / 2.92	0.03 / 0.16	0.067
M030 / upper	88.2	0.00 / 3.55	2.54 / 0.73	0.107
M031 / web	149.7	9.20 / 0.00	0.06 / 0.47	0.456
M032 / upper	38.9	2.17 / 0.00	9.90 / 1.16	0.319
M033 / web	134.5	0.00 / 2.20	0.05 / 0.60	0.132
M034 / upper	1.0	1.40 / 0.00	10.63 / 1.10	0.334
M035 / upper	60.0	1.40 / 0.00	10.44 / 1.07	0.328
M036 / upper	7.0	0.61 / 0.00	3.71 / 0.46	0.119
M037 / web	8.0	0.50 / 0.00	0.20 / 0.34	0.093
M038 / web	32.1	1.13 / 0.00	0.13 / 0.42	0.103
M039 / web	44.4	1.72 / 0.00	0.13 / 0.56	0.135
M040 / web	84.6	3.07 / 0.00	0.06 / 0.98	0.223
M041 / web	128.8	0.00 / 1.30	0.20 / 0.70	0.160
M042 / web	128.8	0.00 / 1.57	0.50 / 0.94	0.252
M043 / web	128.8	0.00 / 1.76	1.25 / 0.93	0.376
M044 / web	128.8	0.00 / 3.16	2.00 / 0.98	0.522
M045 / web	128.8	4.26 / 0.00	2.37 / 0.60	0.609
M046 / web	60.0	0.00 / 0.34	1.86 / 0.01	0.309

Each row reports the controlling sampled load case and finite-element segment within that physical member. Axial and moment components conservatively take the larger end value within that segment; they are not a single connection force vector. The CSV records the governing case, effective buckling length, capacities, shear and torsion for every row.

Worked check: governing member in this truss

M023: HSS 6 x 6 x 1/4; governing case strength B2@0/8. Assumed effective buckling length 72.0 in. Available compression 219.98 kip, tensile yield 235.80 kip, major bending 35.74 kip-ft, minor bending 35.74 kip-ft.

Axial term = 0.003; major-bending term = 0.094; minor-bending term = 0.015. Sum = 0.112. Separate shear/torsion screen = 0.778. Governing metric = 0.778, against the study target 0.900. This conditional screen does not include joint capacity or second-order effects.

T-W — why not make it lighter?

Family	Lighter trial	Saved lb	Global ratio	Global movement in
lower	HSS 4 x 4 x 3/8	50.7	1.110	0.756
upper	HSS 4 x 4 x 3/8	55.9	1.013	0.732
web	HSS 2.5 x 2.5 x 3/16	160.6	0.934	0.742

Each row changes only that family and then re-solves the entire connected building. The illustrated trial is the next lighter catalogue entry by weight; the full search also tested other lighter entries. A ratio above 0.900 or movement above 0.750 inch rejects the trial. A rejection can be caused by a different member because stiffness changes redistribute loads.

Selected T-W weight including 10% connection allowance: 2,020 lb.

Connection actions to carry forward

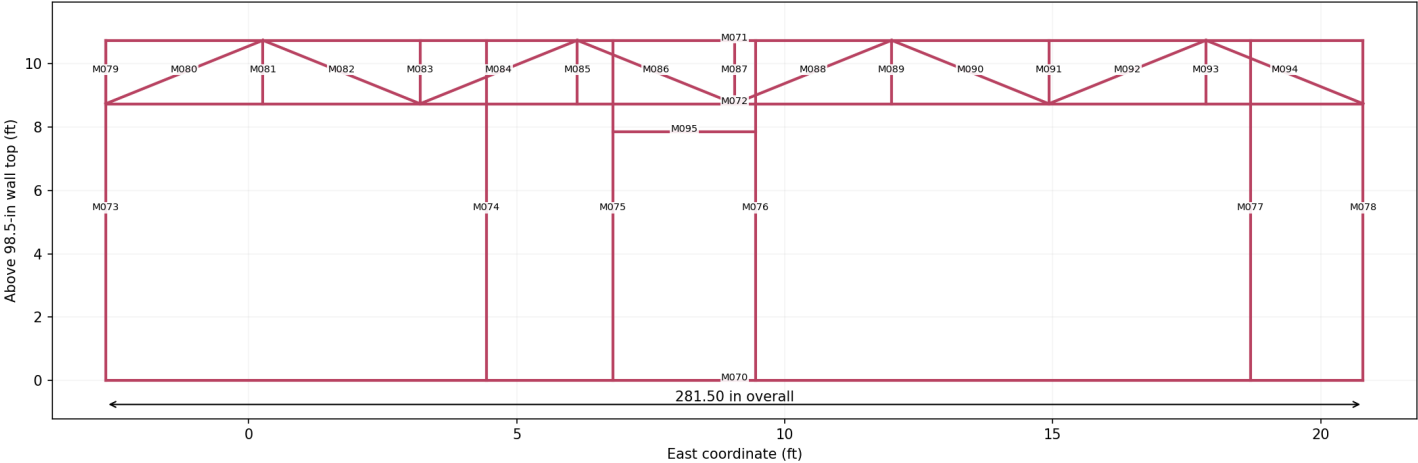
Node	Connected group(s)	Vertical lb	Moment magnitude kip-ft
0	T-S	-612	0.22
13	T-S	+500	0.02
38	O2, T1	+9,190	2.58
51	T1	+1,095	0.08
91	T-SO	+89	0.32
101	B2, O3	-4,211	2.33
106	B3	-3,683	9.23
107	T-N	+374	8.28
145	T-N	+1,970	0.88
171	T-N	+2,322	0.29
172	T-N	+4,174	0.18

These are service connection actions, included to expose the assumed transfer path. They are not a factored connection design schedule. The moment magnitude is for orientation only; connection design must use the signed component vectors and all governing combinations. Welds, gussets, bolts, HSS face yielding and local distortions remain unqualified.

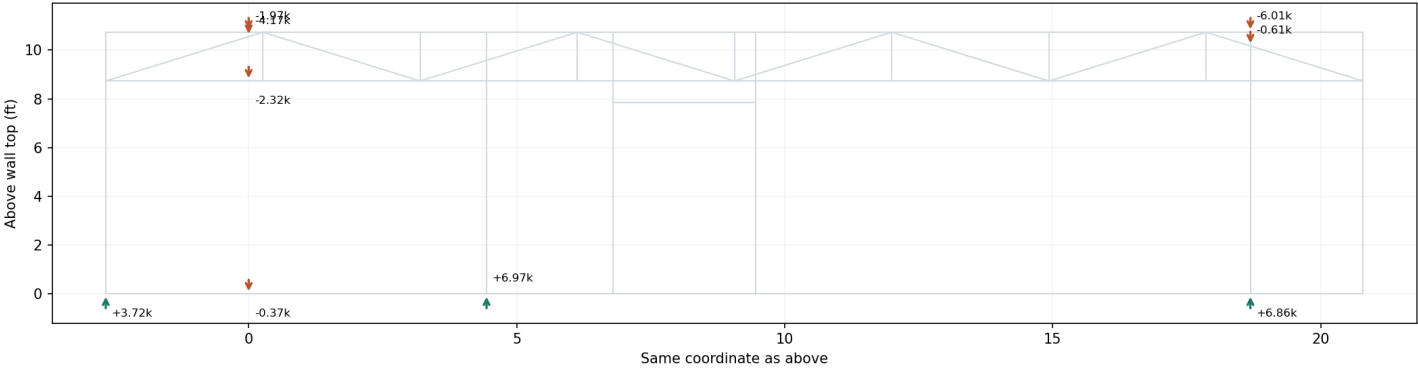
T-N — geometry and load transfer

lower: HSS4x4x0.188 | upper: HSS3x3x0.188 | web: HSS3x3x0.125

T-N — full truss, with physical member IDs



Service storage: 2.07 kip directly applied to this group; arrows = actions from connected framing



Overall analytical span 281.50 in; maximum depth above wall-top line 128.75 in. Physical member IDs match the force table. Member lengths are analytical centerline lengths, not cut lengths. Joints, offsets and weld gaps remain to be detailed.

Role	Candidate	Governing screen	Case
lower	HSS 4 x 4 x 3/16	0.662	strength B2@0/8
upper	HSS 3 x 3 x 3/16	0.436	strength B2@0/8
web	HSS 3 x 3 x 1/8	0.523	strength B2@8/8

The bottom arrows show only vertical interaction with adjacent framing in the no-trolley service case. Distributed direct loads are summarized in the title. Horizontal actions and joint moments also exist; connection-actions.csv contains all six components. A connection cannot be designed from the vertical arrows alone.

The door interrupts triangulation below the upper band. The middle horizontal line and upper band are both in the upper-chord family. Portal-frame joint action around the opening has not been fabricated or verified.

T-N — member force and capacity checks

ID / role	Length in	Comp. / tens. kip	Major / minor M kip-ft	Ratio
M070 / lower	281.5	0.03 / 0.00	7.67 / 0.04	0.662
M071 / upper	281.5	6.10 / 0.00	0.42 / 0.97	0.436
M072 / upper	281.5	0.00 / 9.60	0.82 / 0.29	0.292
M073 / web	128.8	7.62 / 0.00	0.63 / 0.07	0.474
M074 / web	128.8	5.79 / 0.00	0.35 / 0.45	0.420
M075 / web	128.8	2.08 / 0.00	0.18 / 0.21	0.173
M076 / web	128.8	0.00 / 1.43	0.10 / 0.15	0.080
M077 / web	128.8	5.49 / 0.00	0.32 / 1.00	0.523
M078 / web	128.8	3.36 / 0.00	0.08 / 0.21	0.205
M079 / web	24.0	0.38 / 0.00	0.29 / 0.43	0.168
M080 / web	42.6	10.77 / 0.00	0.51 / 0.47	0.424
M081 / web	24.0	0.00 / 2.94	0.77 / 0.45	0.324
M082 / web	42.6	9.44 / 0.00	0.05 / 0.53	0.307
M083 / web	24.0	0.00 / 1.21	0.13 / 0.72	0.211
M084 / web	42.6	0.00 / 4.89	0.52 / 0.07	0.215
M085 / web	24.0	1.39 / 0.00	0.05 / 0.14	0.067
M086 / web	42.6	0.00 / 3.13	0.12 / 0.08	0.099
M087 / web	24.0	0.24 / 0.00	0.03 / 0.21	0.057
M088 / web	42.6	0.90 / 0.00	0.18 / 0.03	0.065
M089 / web	24.0	0.00 / 0.09	0.01 / 0.02	0.009
M090 / web	42.6	0.00 / 0.65	0.01 / 0.01	0.016
M091 / web	24.0	0.00 / 0.06	0.01 / 0.02	0.008
M092 / web	42.6	0.74 / 0.00	0.09 / 0.05	0.046
M093 / web	24.0	0.29 / 0.00	0.15 / 0.13	0.067
M094 / web	42.6	4.19 / 0.00	0.31 / 0.08	0.166
M095 / web	32.0	0.15 / 0.00	0.08 / 0.01	0.022

Each row reports the controlling sampled load case and finite-element segment within that physical member. Axial and moment components conservatively take the larger end value within that segment; they are not a single connection force vector. The CSV records the governing case, effective buckling length, capacities, shear and torsion for every row.

Worked check: governing member in this truss

M070: HSS 4 x 4 x 3/16; governing case strength B2@0/8. Assumed effective buckling length 72.0 in. Available compression 99.18 kip, tensile yield 116.10 kip, major bending 11.66 kip-ft, minor bending 11.66 kip-ft.

Axial term = 0.000; major-bending term = 0.658; minor-bending term = 0.004. Sum = 0.662. Separate shear/torsion screen = 0.187. Governing metric = 0.662, against the study target 0.900. This conditional screen does not include joint capacity or second-order effects.

T-N — why not make it lighter?

Family	Lighter trial	Saved lb	Global ratio	Global movement in
lower	HSS 3 x 3 x 1/4	15.7	0.860	0.811
upper	HSS 2.5 x 2.5 x 3/16	66.1	0.909	0.733
web	HSS 2 x 2 x 3/16	52.7	1.017	0.740

Each row changes only that family and then re-solves the entire connected building. The illustrated trial is the next lighter catalogue entry by weight; the full search also tested other lighter entries. A ratio above 0.900 or movement above 0.750 inch rejects the trial. A rejection can be caused by a different member because stiffness changes redistribute loads.

Selected T-N weight including 10% connection allowance: 1,180 lb.

Connection actions to carry forward

Node	Connected group(s)	Vertical lb	Moment magnitude kip-ft
107	T-W	-374	8.28
145	T-W	-1,970	0.88
171	T-W	-2,322	0.29
172	T-W	-4,174	0.18
189	N2, T-E	+6,856	6.39
229	T-E	-6,012	0.54
256	T-E	-11	0.14
257	T-E	-612	0.03
258	B-WO, W4	+3,719	0.22
262	N1	+6,973	0.64

These are service connection actions, included to expose the assumed transfer path. They are not a factored connection design schedule. The moment magnitude is for orientation only; connection design must use the signed component vectors and all governing combinations. Welds, gussets, bolts, HSS face yielding and local distortions remain unqualified.

B2 and the trolley rails

Rail	Candidate	Actual depth	Screen ratio
B2	W8X31	8.00 in	0.606
T1 lower chord	W8X24	7.93 in	0.446

B2 is a beam and T1 is a truss. B2 uses end bending releases to model shear seats; torsional restraint remains assumed. T1 lower chord remains connected to its posts, diagonal end bays and upper chord. The rest of T1 participates in carrying the trolley and floor loads.

Nine loaded positions were checked on each rail, one trolley at a time. The position label @3/8 means three-eighths of the way from the west end. Moving load is applied on the rail centerline for vertical global analysis. Wheel spacing, unequal wheel loads, longitudinal acceleration, lateral force, flange-local bending, web tension and restrained warping are not resolved by that idealization.

No 1,000 lb lifting rating is established by these results. The actual trolley specification is needed to check flange fit, wheel contact and the relevant hoist standard/dynamic factors. Both the full building load and the moving load must be retained in the final checks. Bolted splices, end stops and stiffeners must preserve trolley travel.

Whole-frame reactions remain important

Column	Service storage reaction, lb
W1	249
W2	12,648
W3	2,260
W4	4,116
S1	85
S2	2,906
N1	7,058
N2	14,783
S3	9,905

These revised reactions supersede the earlier values only for this conditional configuration. Columns remain 4 x 4 x 3/16 HSS trial members. Axial capacity alone does not establish column or footing adequacy. The assumed rigid bases and bracing have not been verified.

Remaining engineering and reproducibility

The geometry still contains architectural-line approximations at joints, rather than fully detailed section-centroid offsets. Upper HSS chord sizes as small as 2 inches may be impractical when connected to the much larger lower chords. Cost, weld access, tube-face capacity and required bracing can make a heavier but simpler truss preferable.

This revision does not optimize geometry, connection weight or purlin profiles. Roof purlin routing is assumed, and the hip-cap rafters and collectors are represented only by loads. The frame is not a complete lateral-force-resisting system. Wind, seismic, rain/ponding, second-order effects, diaphragm actions, uplift connections and foundation capacity must be checked before members are finalized.

The recorded stopping point is local and conditional. It answers “what lighter members survive these explicit gravity screens in this connected model?” It does not answer “what is the lightest safe building?” That latter answer requires the unresolved physical details and site design loads.

Files and commands

`engineer.py` builds the revised connected model, performs the grouped section search and writes `results.json`. `geometry_builder.py` preserves the prior geometry and load routing. `frame_solver.py` now distinguishes W-section strong and weak axes. Run `verify.py` for benchmarks; run `engineer.py`, then `detail.py` for member IDs, CSVs and figures; run `report.py` with ReportLab to produce this document. NumPy, SciPy and Matplotlib are needed for the calculation/figures.

Inputs are frozen under `inputs/`. The prior HSS starting selection is stored as `baseline-selection.json`. Physical member lengths and coordinates are in `physical-members.json`; individual demand/capacity data are in `member-checks.csv`; signed service joint actions are in `connection-actions.csv`; the complete section-search history and lighter trials are in `results.json`. The source ZIP includes the AISC catalogue download used for the W data.

Primary references

[AISC Shapes Database v16.0](#)

[Official AISC spreadsheet downloaded for this study](#)

[Atlas square HSS properties](#)

[AISC specification formula reference, E3 and F2](#)

[AISC discussion of bottom-flange loading and hoist impact](#)

[San Diego adopted codes](#)

The cited formula reference and diagnostic load assumptions do not establish compliance with the currently adopted California code. A licensed structural engineer must reconcile the final design basis, details and applicable standards before construction.